

Fluid-Related Corrosion Investigation

Industrial Manufacturing | Incoming Quality / Supplier Quality Environment

September 2026 • Anonymized real-world case

Confidentiality

This case study is anonymized. Company names, supplier names, internal identifiers, NCR numbers, specifications, material codes and other confidential information have been omitted.

Problem	Rapid corrosion appearing after cleaning, with a distinctive liquid-run-off pattern.
My role	Observation, hypothesis development, practical testing, evidence collection and escalation.
Outcome	The responsible Quality organization confirmed the fluid-related cause and replaced the affected fluid stock/source.
Formal QE ownership	No. I did not own the formal supplier change, quality-standard change or final corrective-action approval.

1. Background

During incoming and pre-production quality activities, I observed a recurring corrosion issue affecting small bearing components and the rings used for connection with a shaft.

The corrosion appeared shortly after cleaning activities and had a distinctive streak-like pattern resembling liquid run-off. The speed of the effect and the absence of an obvious external water source made the process fluid a plausible suspect.

2. Initial Observation

- Visible corrosion streaks appeared after a lunch break on areas that had been cleaned beforehand.
- The effect developed in approximately 30 minutes.

- There was no rain or obvious external water exposure.
- The pattern looked like liquid streaks carrying corrosion, which led me to consider the cleaning/process fluid as a possible cause.

Key decision

I did not treat the suspected fluid as the root cause immediately. I chose to investigate whether the effect could be isolated and reproduced.

3. Investigation Approach

I reviewed the identification of the fluid in use, including its label, supplier and product code. I also checked other available fluids and compared affected and unaffected conditions.

Observe	Hypothesis	Isolate	Test	Escalate
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4. Practical Verification

- I compared an area associated with the suspected fluid against an area not previously exposed to it.
- A different fluid was used as a comparison condition.
- I avoided mixing multiple fluids on the same ring so that the comparison would remain interpretable.
- When the suspected fluid was applied again, the corrosion pattern reappeared.
- The comparison condition did not show the same visible effect.

Evidence

The ability to reproduce the effect with the suspected fluid, while the comparison condition remained unaffected, increased confidence that the fluid was contributing to the corrosion.

5. Evidence and Escalation

I documented the observation and shared photographs, comparisons and my reasoning with the responsible Quality Supplier Lead, Quality Engineer and Incoming Quality team.

The responsible Quality team then repeated the observation together with me. The corrosion reappeared after approximately 10–20 minutes, but only in connection with the specific fluid.

6. Corrective Action and Outcome

The responsible Quality organization confirmed that the specific fluid was contributing to the corrosion issue. The existing stock was replaced and its source/supplier was changed.

Scope of my responsibility

I did not have authority to change the supplier, modify the quality standard or formally approve the corrective action. My responsibility was to identify and investigate the suspected cause and provide evidence to the responsible Quality organization.

After the affected fluid was removed from use, I no longer observed the same corrosion issue associated with that fluid. Other corrosion-related issues occurred later under different conditions, including material/casting-related factors, but those were outside the scope of this case.

7. My Contribution

- Identified an unusual and recurring quality problem rather than treating it as a routine NOK condition.
- Recognized a plausible process-related cause from the timing and physical pattern of the defect.
- Checked relevant process information and fluid identification.
- Compared affected and unaffected conditions and isolated the suspected variable.
- Used a practical verification to reproduce the effect.
- Documented findings and communicated evidence to the responsible Quality functions.
- Supported the subsequent validation performed by the Quality team.

8. Quality Engineering Principles Demonstrated

Principle	How it appeared in the case
Problem recognition	A recurring defect pattern was noticed and treated as an abnormal condition requiring investigation.

Hypothesis development

The fluid was identified as a plausible cause based on timing, appearance and process context.

Variable isolation

Different fluids and exposure conditions were compared without intentionally mixing variables on the same ring.

Evidence-based thinking

The suspected effect was tested and reproduced rather than accepted on assumption.

Verification

The responsible Quality team repeated the observation and confirmed the relationship.

Escalation

Findings were communicated to the functions with authority to decide and implement corrective action.

9. Connection to Formal RCA

At the time of the investigation, I did not formally own an RCA process. However, the investigation followed several principles that I am now formalizing through Quality Engineering development:

Observation	Hypothesis	Test / Evidence	Escalation / Action
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10. Key Learning

From defect detection to problem solving

The key lesson was that a visible defect is only the starting point. The useful question is not only 'what is wrong?' but also 'what changed, what could explain the pattern, can the suspected cause be isolated, and can the effect be reproduced?'

This experience strengthened my interest in moving from hands-on quality inspection toward structured problem solving and Quality Engineering.

Portfolio classification: Real-world experience case • Anonymized / non-confidential • Formal QE ownership: No